An LLM-Augmented Stacked Ensemble for Vietnamese Emotion Recognition on Social Media

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Abstract

Fine-grained emotion recognition from Vietnamese social-media text is a difficult natural language processing problem due to the language's low-resource status. This paper addresses a methodological gap on this problem: existing ensembles typically fuse base-model probabilities with fixed rules or linear stacking and do not condition on the raw sentence, while large language models are used mainly as offline generators or as static prompts. We propose a two-stage framework, evaluated on UIT-VSMEC, a seven-class Vietnamese social-media emotion corpus. In Stage 1, Gemini-2.5-Flash generates and paraphrases minority-class examples via 10-shot prompting. In Stage 2, five heterogeneous base classifiers (PhoBERT, CafeBERT, ViBERT, Logistic Regression, LinearSVC) produce probability distributions, and a QLoRA-fine-tuned Gemma-2-9B-it meta-model consumes the raw sentence and those five distributions jointly. The full framework reaches 70.33% $\pm$ 1.33 accuracy, 70.39% $\pm$ 1.26 weighted F1, and 67.27% $\pm$ 1.25 macro F1 over seeds 42, 131, and 9998. On the same settings, LogReg stacking reaches 67.48% $\pm$ 0.50 weighted F1 and weighted voting reaches 67.91% $\pm$ 1.20.

***Index Terms*—Vietnamese NLP, emotion recognition, data augmentation, large language models, ensemble learning, social media**

I. Introduction

The rapid growth of Vietnamese social-media platforms has produced vast volumes of user-generated text carrying rich emotional signals. Mining these signals through fine-grained Emotion Recognition (ER) yields value for market research, opinion monitoring, and mental health screening, yet ER remains substantially harder than coarse sentiment polarity classification. Sentiment analysis on Vietnamese text—typically framed as a three-class task (positive, negative, neutral)—has benefited from larger annotated corpora, mature pre-trained language models, and relatively clear lexical polarity cues. Hybrid architectures such as PhoBERT combined with CNN-LSTM reach over 93% accuracy on the UIT-VSFC [1]–[3], and several deep-learning baselines report F1 scores above 90% on comparable polarity benchmarks. By contrast, fine-grained emotion recognition assigns each sentence to one of seven distinct affective states, where labels are linguistically subtler and more context-dependent. On UIT-VSMEC, the best reported result from the original benchmark study reaches only 59.74% weighted F1 using CNN with word2vec embeddings [4]. Subsequent monolingual BERT models have narrowed but not closed this divide, indicating that scaling pre-training alone is insufficient for reliable fine-grained Vietnamese emotion classification.

Two obstacles drive this gap. First, Vietnamese is a low-resource language [5], [6]: annotated datasets are scarce. Second, UIT-VSMEC suffers from pronounced class imbalance: the dominant label *Enjoyment* constitutes roughly 28% of the training set, whereas minority emotions such as *Surprise* ($\sim$4.5%) and *Anger* ($\sim$6.9%) are severely underrepresented. Together, these constraints limit the robustness and generalization of individual classifiers.

There is a methodological gap, where existing ensembles typically fuse class probabilities through fixed rules (voting) or linear models (logistic-regression stacking), while large language models (LLMs) are used mainly as offline data generators or as static prompts. We address this gap with a two-stage framework. In Stage 1, Gemini-2.5-Flash [7] performs targeted data augmentation for minority emotion classes using 10-shot prompting. In Stage 2, five heterogeneous base classifiers produce probability estimates; a Gemma-2-9B-it [8] act as the meta learning model is then fine-tuned with parameter-efficient QLoRA [9] on the raw sentence and those distributions jointly.

The main contributions of this paper are:

1) An LLM-based augmentation protocol that generates and paraphrases Vietnamese social-media sentences for minority emotion classes with a controlled 10-shot prompting strategy.
2) A stacked ensemble in which a QLoRA-fine-tuned meta learning model Gemma consumes the raw sentence and five base-model probability distributions.

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II. Related Work

A. Vietnamese Emotion Recognition

Ho et al. [4] introduced UIT-VSMEC, a 6,927-sentence seven-class corpus, establishing a CNN+word2vec baseline of 59.74% weighted F1. Nguyen and Nguyen [10] reached 64.40% through multi-label regression with preprocessing and key-clause extraction. Pre-trained transformers further narrowed the gap: PhoBERT [11] underpins hybrid CNN-LSTM architectures that achieve 93.24% accuracy on the polarity task [3], but the fine-grained ER on informal social-media text remains harder. CafeBERT [12] (XLM-RoBERTa trained on Vietnamese Wikipedia and news) reports 66.12% F1 on VSMEC; ViSoBERT [13] reaches 68.37% weighted F1; and a GRU–CNN–BiLSTM–LSTM ensemble [14] attains 65.79% macro F1. Despite these advances, minority emotions such as Fear, Anger, and Surprise remain difficult to classify and leave room for data-centric and ensemble improvements.

B. LLM-based Augmentation and Ensemble Methods

Ding et al. [15] survey LLM-driven data creation and co-annotation, noting risks of contamination and model collapse. Ubani et al. [16] show zero-shot ChatGPT generation outperforms EDA and back-translation on SST-2 and SNIPS; Whitehouse et al. [17] extend LLM augmentation to cross-lingual settings; Üveges and Ring [18] confirm 10-shot LLM prompting yields macro-F1 gains for underrepresented emotion classes. However, prior work treats the LLM as an offline generator only, or as a static zero-/few-shot classifier, and prompt design for fine-grained Vietnamese emotion categories remains underexplored. On the ensemble side, Thin et al. [19] obtain 94.03% weighted F1 on VSFC [1] via soft-voting over five Vietnamese transformers; Thin et al. [20] further evaluate zero- and few-shot LLM prompting for Vietnamese sentiment. This leaves a methodological gap for Vietnamese emotion recognition: using an LLM both to generate minority-class data and to fuse heterogeneous ensemble probabilities while conditioning on the sentence. We address that gap with the two-stage framework in Section III.

III. Methodology

This study proposes an end-to-end pipeline for processing Vietnamese social media data and recognizing emotions in sentences, as illustrated in Fig. 1. It consists of four stages: LLM data augmentation, text preprocessing, base classifier training with cross-validation stacking, and meta-learning LLM model fine-tuning.

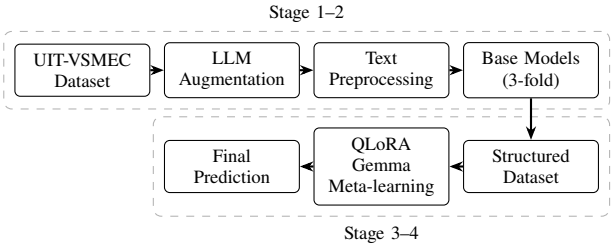


Fig. 1. Proposed pipeline: augmentation and preprocessing feed the base models; their cross validation probabilities form a structured dataset for QLoRA Gemma fine-tuning.

A. LLM-based Data Augmentation

UIT-VSMEC exhibits a 6.4:1 imbalance between the most and least frequent classes. We address this by prompting Gemini-2.5-Flash [7] via the Gemini chat interface to generate additional training examples for the three most underrepresented emotions: Fear, Anger, and Surprise. Two complementary strategies are applied: (i) *generated*—the model synthesizes entirely new Vietnamese sentences conditioned on an emotion label and 10 randomly sampled in-context examples, creating novel scenarios not present in the original corpus; (ii) *paraphrased*—the model rewrites existing annotated sentences with linguistic variation while preserving the target emotion and original semantics. Each class is augmented to a target of 1,500 examples per strategy. Augmentation operates on the raw training split before any preprocessing. Fig. 2 shows the six-stage pipeline.

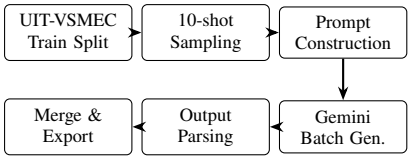


Fig. 2. Synthetic data creation pipeline for minority-class augmentation.

The pipeline operates as follows:

1) **10-shot sampling.** Ten in-class sentences are sampled uniformly at random (with replacement when the class has fewer than ten examples) to form the few-shot context.
2) **Prompt construction.** A system message and structured human message are assembled from the selected template (generated or paraphrased).
3) **Batch generation.** The prepared prompts are sent to Gemini-2.5-Flash in batch. We set generation temperature to 0.9 to encourage varied wording across outputs [21], and apply a frequency penalty of 0.5 so the model is less likely to repeat the same phrases. Each response yields a batch of new Vietnamese sentences for the target emotion.

TABLE I
PROMPT DESIGN COMPARISON: GENERATED VS. PARAPHRASED STRATEGY

	Generated	**Paraphrased**
System role	VN social-media expert; express target emotion only	Gen-Z social-media expert; preserve meaning, tone, intent
Human task	Synthesize NEW sentences for the target emotion (in English and Vietnamese)	Rewrite sentences preserving SAME meaning and emotion
Key constraints	Natural VN style; length mix (5–10 / 10–20 / 20+ words); no other emotions	Semantic equivalence mandatory; no new facts or intensity; preserve negation and intensifiers
Output format	Numbered lines: sequence number, generated sentence, and example reference	

TABLE II
EXAMPLES OF AUGMENTED SENTENCES BY STRATEGY

Strategy	Original	Output	Emotion
Gen.	per không cho anh đi tập đâu nha mất bồ như chơi	Cái cảm giác bị bỏ rơi nó ám ảnh tui ghê lắm.	Fear
Gen.	sao mà mặt boss có thể gian phản diện đến thế nhỉ	Sao nhìn cái mặt của nhân vật phản diện nó gian manh vậy ta.	Surprise
Para.	sợ ngỗng còn hơn sợ chó :)))	Thà sợ chó còn hơn sợ ngỗng :)))	Fear
Para.	vl :)) còn đút tay túi áo ??	Trời đất ơi, còn đút tay vào túi áo nữa hả?	Surprise

1) Prompt Design: Both strategies share the same two-part prompt template structure: a *system message* that constrains style and emotion fidelity, and a *human message* that supplies the emotion label, its Vietnamese description, and 10 few-shot examples. Table I summarises the key differences; Table II shows representative outputs for both strategies (Gen. for Generated and Para. for Paraphrased).

B. Text Preprocessing

Vietnamese social-media text is noisy: users mix emojis, keyboard emoticons, slang, and short forms that carry strong emotional meaning. If these signals are removed too early, models may lose useful cues. Following practices similar to prior Vietnamese social-media emotion work [5], [10], we therefore apply a shared cleaning pipeline to the data after augmentation.

Each sentence is cleaned in four steps. First, a curated dictionary of emoji-to-Vietnamese mappings replaces Unicode emojis with short descriptive phrases (e.g., "vui vẻ" for happy faces and "buồn bã" for sad faces); unmatched emojis are then removed with a Unicode regex filter. Second, text emoticons such as `:))`, `:(`, and `T_T` are mapped to Vietnamese words, with longer patterns matched first to avoid partial replacements. Third, common internet abbreviations are expanded at the word level (e.g., *ko*/*k*/*hok*, all meaning *không*; *dc*/*đc*, meaning *được*). Fourth, URLs and HTML tags are stripped, non-Vietnamese characters are replaced with spaces while diacritics and basic punctuation are kept, repeated punctuation is collapsed, and whitespace is normalized.

After cleaning, duplicate sentences are removed so repeated content does not bias training. Stopwords are intentionally kept. Function words such as *không* (negation) and *rất* (intensifier) often change emotion meaning, and BERT-style encoders can use these tokens through self-attention [11], [12].

C. Base Classifiers

Five heterogeneous base classifiers—PhoBERT-v2 [11], CafeBERT [12], ViBERT [22], Logistic Regression [23], and LinearSVC—are trained with stratified 3-fold cross-validation, each producing probability vectors across the seven emotion classes.

Each model is also refitted on the full training set to produce validation and test probability arrays.

D. Stacked Ensemble with Meta-learning LLM

Weights and structured dataset. Each base model receives one weight, computed on the validation split after that model has been refit on the entire training set. For model i, Acc_i is the fraction of validation sentences whose highest-probability class matches the gold label. The five accuracies are then scaled so the weights sum to one:

$$W_i = \frac{\mathrm{Acc}_i}{\sum_{j=1}^{M} \mathrm{Acc}_j}, \qquad M = 5. \tag{1}$$

A model that classifies more of the validation split correctly therefore takes a larger share of the weighted-average distribution. The five weights are calculated once, before LLM meta-learning training, and the same values are reused in the training, validation, and test prompts.

During meta-learning model training, each training sentence is paired with the seven-class distribution from the fold model that left that sentence out, and W_i is applied to those out-of-fold distributions to form the weighted average. Validation and test prompts use the distributions from the full-training refit, because those sentences were held out of base-model fitting. Every prompt contains the original sentence, the five per-model distributions, and that weighted average, so the meta-model reads the text together with the ensemble signals.

QLoRA fine-tuning. Gemma-2-9B-it [8] serves as the predictor. The model is fine-tuned with QLoRA [9] (4-bit NF4 quantization, rank-16 adapters) using completion-only on the emotion label as the target output. At inference, the model generates the emotion label jointly conditioned on the sentence and the five base-model probability distributions in the structured prompt.

IV. EXPERIMENTS

A. Dataset and Splits

UIT-VSMEC [4] contains 6,927 human-annotated sentences drawn from Vietnamese social-media platforms, covering seven emotion labels: Enjoyment, Sadness, Fear, Anger, Disgust, Surprise, and Other. We used. We set a threshold of 1500 examples, and augmented classes that fall below it to meet the minimum.

B. Reproduction Settings

Table III lists the UIT-VSMEC hyperparameters. The full pipeline is repeated for seeds 42, 131, and 9998. Each seed repeats the stratified folds, base-model training, accuracy-normalized weights, meta-prompts, QLoRA fine-tuning, and evaluation. Ensemble test scores in Table VI are the mean over these three seeds, with the sample standard deviation. Weighted soft voting uses validation-accuracy weights. LogReg stacking selects $C \in \{0.25, 1.0, 4.0\}$ by validation macro F1 (L-BFGS, $\max_\text{iter} = 1000$, seed 123).

TABLE III
UIT-VSMEC HYPERPARAMETERS

Setting	Value
Seeds	42, 131, 9998; each seed is one full pipeline
Ensemble transformers	`vinai/phobert-base-v2`, `uitnlp/CafeBERT`, `FPTAI/vibert-base-cased`; 20 epochs, AdamW 2×10^{-5}, cosine, warmup 0.1, wd 0.01, clip 1.0; batch 128/256, accum. 2, max len 256; patience 3 on macro F1; 3-fold OOF then full-train refit; pipeline seed
Standalone transformers	`uitnlp/CafeBERT`, `uitnlp/visobert` (`use_fast=False`); same optimizer, batch, and length; full-train early stop, patience 5 on macro F1; no OOF; pipeline seed
TF-IDF / linear	1–2 grams, $\min_\text{df} = 2$, sublinear_tf; LogReg and LinearSVC $C = 1.0$, $\text{random_state} =$ pipeline seed, 3 folds; LinearSVC + Platt scaling (inner $cv = 3$)
LogReg stacking	$C \in \{0.25, 1.0, 4.0\}$ by validation macro F1; L-BFGS, $\max_\text{iter} = 1000$, seed 123
QLoRA Gemma	`google/gemma-2-9b-it`; pipeline seed; 4-bit NF4, double quant., bf16; $r = 16$, $\alpha = 32$, dropout 0.05, no bias; targets *q*/*k*/*v*/*o*/gate/up/down proj.; 2 epochs, lr 1×10^{-4}, cosine, 10% warmup, wd 0; batch 16, accum. 2, eval 16, seq 800, completion-only loss; greedy, $\max_\text{new_tokens} = 12$; LoRA inference truncates at 800; zero-shot max length 2048

C. Baselines and Evaluation

We evaluate on the two following experimental conditions. Evaluation metrics are accuracy, weighted F1 (W-F1), and macro F1.

Augmentation ablation. CafeBERT is trained on three versions of the training data—original, paraphrased-augmented, and generated-augmented—and evaluated on the held-out test set using accuracy, weighted F1, and per-class F1 for the three minority emotions (Fear, Anger, Surprise).

Ensemble comparison. Table VI has two blocks. The upper block uses our split, preprocessing, and 692-sentence test set. It compares the stacked QLoRA Gemma meta-model with (i) two conventional fusion baselines on the same five out-of-fold class-probability vectors, without the original sentence: weighted soft voting (accuracy-normalized convex combination, then $\arg\max$) and a multinomial logistic-regression stacking classifier on the concatenated 35-dimensional OOF vector, with C selected on validation macro-F1; (ii) Gemma-2-9B-it zero-shot prompting on the same structured meta-prompts; and (iii) CafeBERT and ViSoBERT rerun on the same training, validation, and test files. The lower block reports previously published UIT-VSMEC scores (CafeBERT and XLM-RoBERTa-Large from VLUE [12], ViSoBERT [13], the MLR pipeline [10], and the GRU–CNN–BiLSTM–LSTM ensemble [14]) for context only; those papers may differ in preprocessing, training, and evaluation, so we do not treat them as same-protocol baselines or as a state-of-the-art comparison. The original CNN+word2vec benchmark [4] (59.74% weighted F1) is omitted from the table for the same reason.

V. Results and Analysis

A. Impact of LLM Augmentation

Table IV reports CafeBERT performance across the three training conditions. Both augmentation strategies yield consistent improvements. The generated data condition produces the largest gains, raising accuracy from 63.44% to 67.03% and weighted F1 from 62.8% to 67.1%. Minority-class gains are particularly pronounced: Anger F1 rises from 38% to 46% (a gain of 8 percentage points), confirming that synthetic generation effectively addresses the most severely imbalanced classes. Fear and Surprise also improve, though more modestly, reflecting their lower initial imbalance. Paraphrasing yields intermediate improvements, suggesting that linguistic variation alone is beneficial but less impactful than generating entirely new examples.

Table V further analyses the quality of the generated sentences. Zero exact and near-duplicate test-set overlap confirms no evaluation leakage in the synthetic corpus. An overall fidelity score of 0.776 shows that generated text is statistically close to original social-media sentences; the KL divergence of 0.164 reflects intentional upsampling of minority classes, not a quality failure. Lower Distinct-2/3 values reveal moderate lexical repetition from 10-shot conditioning, yet generated data still outperforms paraphrasing because novel scenarios expand minority-class coverage more than surface variation alone.

TABLE IV
IMPACT OF LLM AUGMENTATION ON CAFEBERT (UIT-VSMEC TEST SET)

Training Data	Acc.	W-F1	Fear	Anger	Surprise
Original	0.6344	0.628	0.67	0.38	0.60
Paraphrased	0.6676	0.6682	0.67	0.42	0.61
Generated	**0.6703**	**0.671**	**0.68**	**0.46**	**0.61**

TABLE V
QUALITY OF LLM-GENERATED TRAINING SENTENCES (5,010 SYNTHETIC VS. 5,548 ORIGINAL)

Metric	Original	Generated
Overall fidelity score	—	0.776
Distinct-2	0.593	0.451
Distinct-3	0.930	0.786
Mean sentence length (chars)	56.0	51.0
Train–test leakage (exact / near)	—	0 / 0
Emotion distribution KL div.	—	0.164

Fidelity = mean of stat, distribution, label, and correlation similarity.

B. Ensemble Performance

Table VI reports two comparisons. Under our split and preprocessing, the stacked QLoRA system reaches 70.33% ± 1.33 accuracy, 70.39% ± 1.26 weighted F1, and 67.27% ± 1.25

macro F1 (mean ± sample standard deviation over seeds 42, 131, and 9998). The fair transformer baselines are CafeBERT and ViSoBERT retrained on the same files: CafeBERT is the stronger rerun at 64.99% weighted F1, and ViSoBERT reaches 59.66%. Those two reruns are single training runs. The proposed method therefore improves by 5.40 percentage points in weighted F1 over same-protocol CafeBERT and by 10.73 points over same-protocol ViSoBERT. Published CafeBERT (66.12%) and ViSoBERT (68.37%) scores appear only in the lower block. They come from different preprocessing, training, and evaluation settings, so a 70.39% vs. 68.37% gap is not a same-protocol state-of-the-art claim; on this pipeline the ViSoBERT rerun is 59.66% weighted F1. The two

TABLE VI
UIT-VSMEC TEST-SET COMPARISON

Method	Acc.	W-F1	M-F1
Same protocol (ours)			
Proposed (stacked + QLoRA Gemma)	**0.7033 ±0.0133**	**0.7039 ±0.0126**	**0.6727 ±0.0125**
Weighted soft voting (ours)	0.6792 ±0.0123	0.6791 ±0.0120	0.6423 ±0.0106
LogReg stacking meta (ours)	0.6768 ±0.0051	0.6748 ±0.0050	0.6327 ±0.0085
Gemma zero-shot (ours)	0.6618 ±0.0029	0.6572 ±0.0029	0.6172 ±0.0010
CafeBERT, same split (ours)	0.6474	0.6499	0.6142
ViSoBERT, same split (ours)	0.5997	0.5966	0.5604
Published scores (different protocol)			
MLR + preprocessing [10]	0.6436	0.6440	—
CafeBERT [12]	—	0.6612	—
ViSoBERT [13]	0.6810	0.6837	—
XLM-RoBERTa-Large [12]	0.6137	0.6020	—
GRU–CNN–BiLSTM–LSTM [14]	—	—	0.6579

Note. Upper block: same split, preprocessing, and 692-sentence test set. The four ensemble rows are mean ± sample standard deviation over seeds 42, 131, and 9998. CafeBERT and ViSoBERT rows are single runs. Lower block: previously published scores under other protocols; not used to claim state-of-the-art. W-F1: weighted F1; M-F1: macro F1; —: not reported.

conventional fusion rows compare the QLoRA meta-learner with ordinary stacking on the same five out-of-fold class-probability vectors. Weighted soft voting applies *fixed* fusion of those vectors (already supplied to Gemma as a consensus summary). LogReg stacking trains a *linear* meta-classifier on the concatenated 35-dimensional OOF vector, without the sentence, with C selected on validation macro-F1. Zero-shot Gemma uses the same structured meta-prompts without supervised fine-tuning. QLoRA Gemma attains 70.39% weighted F1, against 67.91% for voting (+2.48) and 67.48% for LogReg (+2.91). The macro-F1 relative to LogReg is +4.00 (67.27% vs. 63.27%). All three stacking systems use the same five base models and the same OOF signals; QLoRA Gemma additionally conditions on the sentence. This comparison does not evaluate non-linear heads such as MLP or XGBoost.

A 4.67 percentage-point gain in weighted F1 over zero-shot Gemma (70.39% vs. 65.72%) shows that QLoRA fine-tuning on the structured dataset helps relative to zero-shot inference. The best same-protocol single transformer (CafeBERT, 64.99% weighted F1) trails the full ensemble by 5.40 points, showing that stacking heterogeneous BERT and TF-IDF signals goes beyond any individual classifier on this pipeline. The CafeBERT augmentation ablation in Table IV (67.10% weighted F1 on generated data) is a separate training-data study and is not the same-protocol CafeBERT baseline in Table VI.

VI. CONCLUSION

A. Summary

This paper proposed a two-stage framework for Vietnamese social-media emotion recognition, addressing a methodological gap in how LLMs and ensembles are combined on this task.

In Stage 1, few-shot LLM augmentation consistently improved CafeBERT's performance across all three under-represented emotion categories. Generated synthetic data—new sentences conditioned on 10 in-context examples—outperformed paraphrasing, raising weighted F1 from 62.8% to 67.1% and increasing Anger F1 by 8 percentage points (38% to 46%). A fidelity analysis confirmed that the synthetic corpus is statistically plausible (overall fidelity score 0.776, zero test-set leakage), establishing that new scenario coverage matters more than mere surface variation.

In Stage 2, stacking heterogeneous BERT and TF-IDF signals through a QLoRA-fine-tuned meta-model reached 70.33% ± 1.33 accuracy, 70.39% ± 1.26 weighted F1, and 67.27% ± 1.25 macro F1 on UIT-VSMEC under our split and preprocessing (mean ± sample standard deviation over seeds 42, 131, and 9998). Relative to same-protocol reruns, this is 5.40 percentage points in weighted F1 above CafeBERT (64.99%) and 10.73 points above ViSoBERT (59.66%). Published transformer scores are not used as a state-of-the-art comparison because they follow different protocols. The full ensemble surpassed zero-shot Gemma prompting by 4.67 percentage points in weighted F1 (70.39% vs. 65.72%). On the same out-of-fold probability features, weighted soft voting reached 67.91% ± 1.20 weighted F1 and logistic-regression stacking reached 67.48% ± 0.50. Stage 1 uses the LLM to generate minority-class data; Stage 2 uses a QLoRA meta-learner that jointly consumes the sentence and the five base-model probability distributions.

B. Future Work

Future work will explore the following directions:

1) **Cross-dataset and cross-domain generalization.** Future work should evaluate the framework on related Vietnamese NLP tasks (aspect-level sentiment, hate-speech detection, multi-label corpora) to verify that LLM augmentation and stacked ensembling generalize beyond fine-grained emotion recognition.
2) **Automated quality validation for synthetic data.** The moderate Distinct-2/3 scores observed in Table V call for automated filtering before synthetic examples enter training: candidate filters include perplexity thresholds, emotion-consistency scoring via a held-out classifier, semantic-similarity caps relative to near-duplicates, and LLM-as-judge fluency checks. The goal is to retain

minority-class coverage gains while improving lexical diversity and reducing formulaic patterns.

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